

SECTION A

1 (short)

(a) A mass m_A of metal at a temperature T_A is put into thermal contact with another mass m_B of the same material at a temperature T_B . Derive an expression for the temperature of the masses T_F when thermal equilibrium is reached. [3]

(b) Alternatively, a heat engine could be connected between the masses (initially at their original temperatures). In this case show that, if the maximum possible work is to be delivered by the engine, then the following identity must hold:

$$m_A \frac{dT_H}{T_H} + m_B \frac{dT_C}{T_C} = 0$$

where T_H and T_C are the instantaneous temperatures of m_A and m_B respectively. Hence, or otherwise, derive an expression for the final temperature of the masses in terms of the initial temperatures T_A , T_B and r , where $r = m_A/(m_A + m_B)$. [7]

2 (short)

(a) Use the first law of thermodynamics, in the form $dq = du + dw$ to show that for a reversible process

$$Tds = du + pdv = dh - vdp$$

and explain why this expression may also be used between equilibrium states joined by irreversible processes. [3]

(b) On a $T - s$ diagram, draw, clearly label and justify the trends for the following processes involving a perfect gas:

- (i) constant enthalpy;
- (ii) constant pressure;
- (iii) constant density;
- (iv) reversible and adiabatic. [7]